

Advanced Econometrics – Lab 03

1 Binary choice models

Exercise 1

Estimate the probit model using data `oscar.csv` of the form

$$winner_i^* = \beta_0 + \beta_1 nominations_i + \beta_2 gglobes_i + \varepsilon_i. \quad (1)$$

The dependent variable *winner* is a dummy variable (= 1 if the movie won the best picture Oscar, = 0 otherwise). We are going to use the number of nominations and the number of Golden Globes won by the movie as explanatory variables.

- Are β_1 and β_2 jointly significant?
- Calculate marginal effects for the average observation and for your own defined observation.
- Calculate R^2 statistics and interpret them.
- Perform the model specification test (linktest).
- Perform the Hosmer-Lemeshow test.
- Perform the Osious-Rojek test
- Verify a hypothesis that $\beta_2 = 0$ using likelihood ratio test.

Exercise 2 (based on Example 16.3.3 from [1])

The Olympic Games are a subject of great interest to the global community. Rightly or wrongly the attention focuses on the number of medals won by each country. Andrew Bernard and Meghan Busse¹ examined the effect of a country's economic resources on the number of medals won. The data are in the file *olympics.csv*.²

- Estimate the linear probability model for a dummy variable *ifgold* equal to one if a country won at least one gold medal at the Olympic Games in Montreal in 1976.

$$ifgold_i = \beta_0 + \beta_1 \ln(POP)_i + \beta_2 \ln(GDP)_i + \beta_3 host_i + \beta_4 planned_i + \varepsilon_i. \quad (2)$$

- Perform specification test.
- Check whether the residuals are homoscedastic.
- Use White's estimator for the variance-covariance matrix.
- Explain why shouldn't we use the linear probability model and what model should be used instead.

¹"Who Wins the Olympic Games: Economic Resources and Medal Totals", *The Review of Economics and Statistics*, 2004, 86(1), 413-417.

²[1] page 440.

Exercise 3 (based on Exercise 16.12 from [1])

Predicting U.S. presidential election outcomes is a weekly event in the year before an election. In the 2000 election, Republican George W. Bush defeated Democrat Al Gore, and in 2004 George Bush defeated Democrat John F. Kerry. The data file *vote2.csv* contains data on these two elections. By state and for the 2 years we report the dummy variable $dem = 1$ if the popular vote favoured the democratic candidate, $income$ = state median income, hs = percentage of the population with at least a high school degree, BA = percentage of the population with at least a bachelor's degree, $density$ = population per square mile, and $region = 3$ for the southwest, 2 for the south, and 1 otherwise.

1. Consider the probit model

$$dem_i^* = \beta_0 + \beta_1 income_i + \beta_2 income_i^2 + \beta_3 region3_i + \varepsilon_i \quad (3)$$

- i. Calculate analytically the marginal effect of the *income* on the probability of voting on a Democrats' candidate.
 - ii. Use R to calculate the marginal effect numerically.
 - iii. Should we add a covariate to the model (3)? If so, what are the implications for the model (3)?
 - iv. Test the hypothesis $H_0 : \beta_1 = \beta_2 = 0$ using the Wald test.
 - v. Test the hypothesis $H_0 : \beta_1 = \beta_2 = 0$ using the Likelihood Ratio test.
2. Estimate the probit model

$$dem_i^* = \beta_0 + \beta_1 income_i + \beta_2 income_i^2 + \beta_3 region3_i + \beta_4 hs_i + \beta_5 ba_i + \beta_6 density_i + \xi_i \quad (4)$$

Discuss the warning message.

Exercise 4 (based on data and example from [2])

The dataset *womenwk.csv* contains information on 2,000 women, 657 of whom are not recorded as wage earners. The indicator variable **work** is set to zero for the nonworking and to one for those reporting positive wages. We fit a probit model of the decision to work depending on the woman's age, marital status, number of children, and level of education.³

- a) Estimate the probit model with the above-mentioned covariates.
- b) Calculate marginal effects.
- c) Calculate R^2 statistics and interpret them.
- d) Perform the model specification test (linktest).
- e) Perform the Hosmer-Lemeshow test.

³[2] p.251-252.

Exercise 5

An econometrician estimated the probit model for the probability of buying a brand-new car

$$newcar_i^* = \beta_0 + \beta_1 age_i + \beta_2 age_i^2 + \beta_3 male_i + \varepsilon_i. \quad (5)$$

Calculate and interpret the effect of *age* on the probability.

Hint: In the probit model

$$Pr(y_i) = \Phi(\mathbf{x}_i' \boldsymbol{\beta}), \quad (6)$$

in which $\Phi(\cdot)$ is the cumulative density function of the standard normal distribution.

Exercise 6

Suppose you buy a weighted coin in a magic shop. It looks like a typical coin with heads and tails, but the coin is not necessarily fair. You flip the coin eight times and observe the sequence HHHHTHTT. What would be a good estimate for p , the probability of heads, based on this sequence? **Find maximum likelihood estimate of p .**

Let $X_i \sim \text{Bern}(p)$, where $X_i = 1$ indicates heads. We will assume that the flips are independent.

$$\begin{aligned} & P(X_1 = 1, X_2 = 1, X_3 = 1, X_4 = 0, X_5 = 1, X_6 = 0, X_7 = 1, X_8 = 0) \\ &= P(X_1 = 1)P(X_2 = 1)P(X_3 = 1) \cdots P(X_8 = 0) \\ &= p^5(1 - p)^3 \end{aligned}$$

Exercise 7*

Let's assume a simple model for Oscar movies

$$oscar_i = \beta_0 + \beta_1 nominations_i + \beta_2 gglobes_i + \varepsilon_i \quad (7)$$

Probability of the success is then

$$Prob(oscar_i = y_i) = [\Phi(\beta_0 + \beta_1 nom_i + \beta_2 gg_i)]^{y_i} [1 - \Phi(\beta_0 + \beta_1 nom_i + \beta_2 gg_i)]^{1-y_i} \quad (8)$$

The likelihood function is then

$$L(\beta_0, \beta_1) = \prod_{i=1}^N [\Phi(\beta_0 + \beta_1 nom_i + \beta_2 gg_i)]^{y_i} [1 - \Phi(\beta_0 + \beta_1 nom_i + \beta_2 gg_i)]^{1-y_i} \quad (9)$$

And the log-likelihood function

$$\ln L(\beta_0, \beta_1) = \sum_{i=1}^N y_i \Phi(\beta_0 + \beta_1 nom_i + \beta_2 gg_i) + \sum_{i=1}^N (1 - y_i) [1 - \Phi(\beta_0 + \beta_1 nom_i + \beta_2 gg_i)] \quad (10)$$

Now, we have to maximise this function with regard to β_0 , β_1 and β_2 . **Let's do it in R!**

Literature

[1] R. Carter Hill, William E. Griffiths and Guay C. Lim, **Principles of Econometrics**, John Wiley & Sons.

[2] Christopher F. Baum, **An Introduction to Modern Econometrics Using Stata**, Stata Press, 2006.